Rank-maximality probabilities under uncertain preferences

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10 October 2026

Abstract

For a specified house allocation, we classify the probability of rank maximality under uncertain strict preferences. Both independent rational lotteries and independent uniform tie refinements give an $\mathsf{FP}^{\#\mathsf{P}}$-complete exact evaluation problem and a PP-complete threshold problem under polynomial-time Turing reductions. The lower bound uses only two-house lists: one exact probability reveals the independent-set count as a sum of integer digits. Explicit joint lotteries admit polynomial-time evaluation and comparison.

Applicants rank their acceptable houses; houses have no preferences. Acceptability is fixed. A feasible matching may leave applicants or houses unmatched. Its rank vector counts the assigned rank-one edges, then the rank-two edges, and so on, padded to the largest acceptable-list length. It is *rank-maximal* when this vector is lexicographically maximal over all feasible matchings. Cardinality is not maximized beforehand. For a fixed feasible matching M, write $\eta(M)$ for the probability of this event. The survey [1, §3.1.2] proposes studying rank-maximal matchings under uncertainty; catalogue entry C78-464 specifies the probability-classification problem considered here.

Theorem 1. *For each of the following independent uncertainty models, exact rational evaluation of $\eta(M)$ is $\mathsf{FP}^{\#\mathsf{P}}$-complete and deciding $\eta(M) \geq q$ for a rational input q is PP-complete, under polynomial-time Turing reductions:*

1. *each applicant has an explicitly listed rational lottery over strict orders;*
2. *each applicant has a weak order, independently and uniformly refined within every tie.*

Hardness holds with at most two acceptable houses per applicant and, respectively, fair two-outcome nontrivial lotteries or nontrivial ties of size two. For an explicitly listed lottery over whole profiles, both tasks are polynomial-time.

The two-choice construction

Let $G = (V, E)$ be a finite simple graph, with n vertices and m edges, and let $t \geq 1$. For each $v \in V$, create houses h_v, k_v, a variable applicant a_v accepting h_v, k_v, and a guard b_v accepting only k_v. For each edge $\{u, v\}$, create t additional applicants accepting h_u, h_v. Every two-house list is a single tie of size two, refined independently and fairly. The same distribution is an explicit lottery over its two strict orders.

The reference matching M assigns a_v to h_v and b_v to k_v, and leaves all edge applicants unmatched. It occupies every house. For a realized profile, let S contain the vertices whose variable applicants rank k_v first. The reference rank vector is $(2n - |S|, |S|)$ when $n > 0$.

Lemma 1. *The reference matching is rank-maximal exactly when every edge applicant's first choice is h_v with $v \notin S$.*

Proof. If an edge applicant first chooses h_v with $v \in S$, remove (a_v, h_v) from M and assign that previously unmatched applicant to h_v. This feasible matching gains one rank-one edge and loses one rank-two edge, so it improves the rank vector.

Conversely, suppose all edge applicants first choose inactive homes. Every first-choice house belongs to

$$\{k_v : v \in V\} \ \cup \ \{h_v : v \notin S\}.$$

Thus any matching has at most $2n - |S|$ rank-one edges, since assigned houses are distinct. The reference attains this bound. A competitor attaining it has at most $2n$ assigned edges in total and hence at most $|S|$ rank-two edges. All acceptable lists have length at most two, so no later coordinate remains to compare. □

One exact query counts independent sets

By Lemma 1, a successful profile must have S independent. For such an S, let $d(S)$ count the graph edges incident to S. Each of their t copies has exactly one successful order; a copy of an edge disjoint from S has two. The variable orders are fixed by S. The successful-profile count N_t therefore satisfies

$$N_t = \sum_{S \in \mathcal{I}(G)} 2^{t(m-d(S))}, \qquad \eta_t = \frac{N_t}{2^{n+tm}}. \tag{1}$$

Here $\mathcal{I}(G)$ includes the empty independent set.

Choose $t = n + 1$ and put $B = 2^{n+1}$. If c_j counts the independent sets with $d(S) = j$, then

$$N_t = \sum_{j=0}^{m} c_j B^{m-j}, \qquad 0 \le c_j \le 2^n < B.$$

These coefficients are exactly the base-B digits of N_t. One exact probability query gives N_t after multiplication by $2^{n+(n+1)m}$; summing its digits gives $|\mathcal{I}(G)|$. The constructed instance has $2n + (n + 1)m$ applicants and $2n$ houses. All integers in the extraction have polynomial bit length, so construction and digit extraction take polynomial time. Independent-set counting is #P-complete under polynomial-time Turing reductions [2, §1, problem 2; §2, reduction 2]. This proves the exact lower bound for both encodings.

For the threshold lower bound, put $D = 2^{n+(n+1)m}$. Queries at $q = k/D$ determine whether $N_t \ge k$. Binary search recovers N_t in $O(\log(D+1))$ queries, after which digit extraction recovers the independent-set count. A #P oracle suffices to decide every PP language, so the threshold problem is PP-hard under the stated Turing reductions. This does not assert many-one hardness.

Isolated vertices need no modification: both variable orders contribute to the corresponding coefficient. When $m = 0$, $N_t = 2^n$ and $\eta_t = 1$, and the single digit counts all subsets. The empty graph gives probability one and one independent set.

Verification and upper bounds

For a deterministic profile with a applicants and maximum list length $R > 0$, assign an edge of rank r the integer weight $(a + 1)^{R-r}$. The first differing rank coordinate dominates all later coordinates: their total possible loss is at most

$$a \sum_{r=j+1}^{R} (a+1)^{R-r} = (a+1)^{R-j} - 1.$$

Maximum-weight bipartite matching, permitting unmatched vertices, therefore verifies rank maximality in polynomial bit time. The case $R = 0$ is immediate.

For rational independent lotteries, choose a common denominator D_i for each applicant and represent each listed mass by its integer number of tickets. Guess one binary ticket per applicant, reject out-of-range tickets, and accept precisely the profiles passing the verifier. The accepting-branch count is $N = D\eta(M)$, where $D = \prod_i D_i$ is known and has polynomial bit length. For uniform refinements, a ticket indexes a permutation in each tie, giving $D = \prod_T |T|!$ and the same argument. Consequently exact evaluation belongs to $\mathsf{FP}^{\#\mathsf{P}}$. For $q = u/v \in (0, 1]$ with $v > 0$, comparison is equivalent to $vN > uD - 1$. The count vN remains in #P: add a bounded ticket guess with v choices. A #P count exceeding a polynomial-time computable integer is a PP predicate [3, Definition 2.3]; thresholds outside $(0, 1]$ are immediate. For explicitly listed joint support, run the verifier on each profile and add the successful masses. This proves all upper bounds and Theorem 1.

Attribution. The earlier GPT 6 Astra Ultra catalogue attempt already gives the verifier, counting upper bounds and joint-lottery algorithm; its two independent-model lower bounds remain unresolved. The new contribution is the gadget, (1), and digit extraction.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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